

PENZA, Russian Federation

WMO: 279620

Lat: 53.1228N

Lon: 45.0189E

Elev: 565

StdP: 14.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-10.7	-21.9	1.7	-16.5	-15.8	2.4	-10.0	24.3	25.9	22.4	24.7	5.1	220	0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.7	67.9	86.0	66.9	82.6	65.5	70.6	83.2	69.1	81.4	67.5	79.2	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	99.4	75.6	64.8	93.7	73.8	63.3	89.0	72.5	34.8	83.0	33.5	81.5	32.3	79.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.8	20.3	18.7	-23.0	93.5	6.1	3.9	-27.4	96.3	-31.0	98.6	-34.4	100.8	-38.8	103.7		
(5)			-22.3	73.9	6.6	2.1	-27.0	75.4	-30.9	76.6	-34.5	77.8	-39.3	79.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	15.0	15.9	25.7	44.4	57.4	64.2	68.7	65.5	55.2	42.4	29.6	19.7		
	DBStd	21.60	12.68	12.52	9.37	9.61	8.74	7.61	6.43	7.31	8.02	8.43	9.91	12.24		
	HDD50	4898	1086	954	753	221	30	3	0	36	263	612	940			
	HDD65	8729	1551	1374	1218	618	265	106	36	85	308	701	1062	1405		
	CDD50	2019	0	0	0	54	259	428	579	479	193	27	0	0		
	CDD65	375	0	0	0	1	29	82	150	99	14	0	0	0		
	CDH74	4017	0	0	0	17	368	888	1555	1042	147	0	0	0		
	CDH80	1358	0	0	0	0	83	288	576	384	27	0	0	0		
	WSAvg	8.6	9.1	9.8	9.6	8.6	9.1	8.0	6.8	7.3	7.9	8.9	9.3	9.4		
	PrecAvg	21.3	1.6	1.2	1.2	1.4	1.6	2.3	2.6	2.2	2.0	1.9	1.9	1.7		
(15) Precipitation	PrecMax	28.3	4.3	3.3	3.8	3.6	3.7	5.5	7.9	4.5	6.1	4.2	4.3	3.3		
	PrecMin	13.1	0.1	0.0	0.1	0.3	0.2	0.5	0.5	0.0	0.0	0.0	0.2	0.6		
	PrecStd	4.3	0.9	0.8	0.7	0.8	0.9	1.3	1.5	1.1	1.4	1.1	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	37.1	51.9	76.7	86.3	93.3	95.4	95.3	83.3	68.4	52.7	42.4		
		MCWB	36.8	35.2	44.9	55.2	63.0	66.9	67.7	68.7	61.5	54.9	49.5	40.8		
	2%	DB	34.2	35.2	44.2	71.3	82.3	87.0	89.7	88.7	78.1	62.9	48.9	37.3		
		MCWB	33.3	33.8	38.7	54.1	61.8	67.0	68.4	67.9	60.1	52.8	47.0	35.9		
	5%	DB	33.2	33.8	40.0	66.0	78.5	82.8	86.1	84.1	74.1	58.4	44.9	35.3		
		MCWB	32.4	32.8	36.3	51.7	60.8	65.6	67.9	66.2	59.4	52.0	43.2	34.4		
	10%	DB	31.8	32.3	37.2	61.0	74.2	79.0	82.5	79.5	69.6	54.7	41.4	33.7		
		MCWB	31.0	31.4	34.5	49.0	58.7	64.5	67.1	64.6	57.4	49.6	40.1	32.9		
	0.4%	WB	36.5	35.6	44.5	58.6	66.3	71.4	73.2	72.4	65.4	58.4	50.3	41.0		
		MCDB	36.9	36.3	51.3	69.4	80.2	84.5	85.3	85.8	74.1	62.7	51.8	41.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.6	33.9	39.6	55.7	63.8	69.3	71.1	69.7	62.7	55.2	46.9	36.0		
		MCDB	33.9	34.5	43.0	66.0	77.7	81.5	83.6	82.9	73.4	59.8	48.6	36.8		
	5%	WB	32.3	32.8	36.6	52.9	62.0	67.6	69.6	67.8	60.7	52.4	43.4	34.2		
		MCDB	32.9	33.5	39.3	64.1	75.7	79.1	81.9	80.8	70.4	57.4	44.6	34.8		
	10%	WB	30.7	31.3	34.6	50.3	59.9	65.6	68.1	66.1	58.6	50.1	40.2	32.8		
		MCDB	31.5	32.3	36.6	60.1	71.3	76.3	79.7	77.4	67.7	54.0	41.2	33.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	13.1	14.0	18.6	21.5	20.5	20.7	21.1	18.6	13.4	9.1	9.9		
		MCDBR	7.8	8.7	15.0	26.1	26.9	25.9	26.6	27.6	25.2	20.0	12.0	6.8		
		MCWBR	8.0	8.2	10.4	12.8	11.9	10.6	10.6	11.0	11.5	12.3	10.1	6.6		
		MCDBR	7.5	8.9	12.6	23.7	24.0	22.5	23.0	24.0	22.1	17.5	10.5	7.0		
	5% WB	MCWBR	7.8	8.6	9.3	13.0	11.6	10.6	10.7	10.8	11.7	12.0	9.6	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.293	0.326	0.359	0.426	0.393	0.387	0.404	0.427	0.377	0.354	0.326	0.296		
	taud		2.099	2.034	2.035	2.090	2.310	2.361	2.317	2.240	2.357	2.340	2.240	2.107		
	Ebn at Noon		207	236	256	253	272	274	266	252	252	230	196	179		
	Edh at Noon		23	34	42	45	38	37	38	39	30	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		278	595	1080	1372	1767	1857	1861	1528	1003	499	210	159		
	RadStd		53	95	139	134	153	170	160	134	137	75	43	44		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		+1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+275	N/A	

Nomenclature: See separate page